Probability Theory

Lecture Notes

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Course Probability Theory

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Lecture 01

Introduction to Probability

Date: Wednesday, 05 August 2026
Instructor: Dr. Nishant Chandgotia

1.1 Motivation and Some Problems

Question 1. Sleeping Beauty Problem

A princess is put into a deep sleep by a powerful drug administered by a villain. The drug ensures that she will remain asleep for two years, except for possible brief awakenings determined by the outcome of a fair coin toss.
The villain tosses a fair coin.

- If the coin lands **Heads**, the princess is awakened *once*, briefly, at the end of the **first year**. After this awakening, her memory of having awakened is completely erased, and she is put back to sleep until the end of the second year.
- If the coin lands **Tails**, the princess is not awakened after the first year. Instead, she is awakened *once*, briefly, at the end of the **second year**.

Whenever the princess awakens, she has no information about the current date or the passage of time. Furthermore, if she had awakened previously, all memory of that awakening has been erased. Consequently, upon waking, she cannot distinguish between the two possible awakening scenarios.
Suppose that the princess wakes up and is immediately asked the following question:

"What probability should you assign to the event that the coin landed Heads?"

What should the princess answer?

Solution:

1. $\frac{1}{2}$ No information exchanged so probability should be $\frac{1}{2}$, If repeated indipendently a milliion times, half chances each.
2. $\frac{2}{3}$ There are possibilities of this. Proportion of waking during Heads is $\frac{2}{3}$,and $\ldots \frac{1}{3}$
3. $\frac{3}{4}$

Villain tosses once, H $\rightarrow$Villain tosses again, Two Possibilities H and T, so,

- HH $\rightarrow \frac{1}{4}$
- HT $\rightarrow \frac{1}{4}$

- T $\rightarrow \frac{1}{2}$

Important

All the solutions above are wrong. The Above problem has no solution, Because it is ill posed, and this ill posedness is with respect to current Framework.

Question 2. Monty Hall Problem

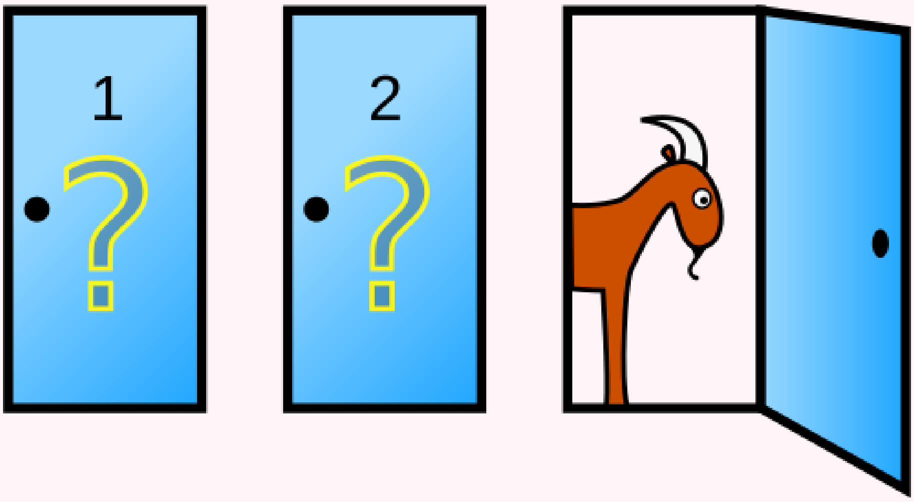

Figure 1: Monty Hall Problem

There are three doors: one hides a car and the other two hide goats. After you choose one door say A, the host opens another door revealing a goat.

1. Question: Should you switch?
2. Probability car behind C > Probability car behind A.

Solution: Yes, we should change our decision,

Initially, the probability that your chosen door contains the car is $P(\text{chosen door}) = \frac{1}{3}$, while the probability that the car is behind one of the other two doors is $P(\text{other two doors}) = \frac{2}{3}$.

Since the host always reveals a goat, this entire probability concentrates on the remaining unopened door. Therefore, $P(\text{Win by staying}) = \frac{1}{3}, \quad P(\text{Win by switching}) = \frac{2}{3}$. Hence, switching gives twice the probability of winning compared to staying.

Remark

The Above question (Monty Hall Problem) is not ill posed.

Summary

- Sleeping Beauty Problem
- Monty Hall Problem

End of Lecture

Probability Theory

Lecture 02

Introduction to Probability

Date: Friday, 07 August 2026
Instructor: Dr. Nishant Chandgotia

2.1 Some Introductory Idea

Suppose we toss a fair coin n-times. $HTH\ HTTH\ \cdots\ T$

What is the Probability of the above sequence ? Its- $\sim \frac{1}{2^n}$

If we have all sequences with k-heads then $P(k\text{-heads}) = \frac{\binom{n}{k}}{2^n}$, and it will be maximum when $k = \frac{n}{2}$.

If $x_1, x_2, \ldots, x_k$ are randomly independently chosen, then $\sum_{i=1}^{k} x_i \sim \frac{r+1}{2} \cdot k$, if they are chosen from the uniform distribution on $\{1, 2, \ldots, r\}$.

Now Stirling's Formula, $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \quad$ for very large n.

Now,

$$\frac{P(\text{number of heads } = k)}{P(\text{number of heads } = k+1)} = \frac{k+1}{n-k}.$$

So, $\frac{k+1}{n-k} > 1 \iff k > \frac{n-1}{2}$, and $\frac{k+1}{n-k} < 1 \iff k < \frac{n-1}{2}$.

As $\binom{n}{k}$ is maximum when $k = \frac{n}{2}$, therefore, $\frac{\binom{n}{n/2}}{2^n} \sim \frac{2}{\sqrt{2\pi n}}$ (By using Stirling's Formula). It would be unjustified to say #Heads $= \frac{n}{2}$ with high probability.

Suppose $k = \alpha n, \qquad \alpha \neq \frac{1}{2}$.

Then $\frac{\binom{n}{k}}{2^n} \sim \frac{1}{\sqrt{2\pi n}} e^{n(-\alpha \log \alpha - (1-\alpha)\log(1-\alpha) - \log 2)}$.

Let, $H(\alpha) = -\alpha \log \alpha - (1-\alpha)\log(1-\alpha)$, be Boltzman Shanon Entropy (Mathematical theory of Communications.

Hence,

$$\frac{\binom{n}{k}}{2^n} \sim \frac{1}{\sqrt{2\pi n}} e^{n\left(H(\alpha) - H\left(\frac{1}{2}\right)\right)}.$$

Now, $H(\alpha) \leq H\left(\frac{1}{2}\right)$, by using Jensen's inequality as log is a concave function.

Equivalently, $-\alpha \log \alpha - (1-\alpha)\log(1-\alpha) < \ln 2$.

Therefore, $\frac{\binom{n}{k}}{2^n} \longrightarrow 0$ exponentially.

Now,

$$P(\#\text{Heads} \in ((\alpha-\varepsilon)n, (\alpha+\varepsilon)n)) = \sum_{k=(\alpha-\varepsilon)n}^{(\alpha+\varepsilon)n} \frac{\binom{n}{k}}{2^n}. \qquad \text{, where} \varepsilon > 0, \qquad \frac{1}{2} \notin (\alpha-\varepsilon, \alpha+\varepsilon).$$

Also,

$$\frac{2n\varepsilon}{2^n}\binom{n}{(\alpha-\varepsilon)n} \leq \cdots \leq \frac{2n\varepsilon}{2^n}\binom{n}{(\alpha+\varepsilon)n}.$$

Therefore, the asymptotic bounds are $\frac{2n\varepsilon}{\sqrt{2\pi n}} e^{n\left(H(\alpha-\varepsilon)-H\left(\frac{1}{2}\right)\right)}$ and $\frac{2n\varepsilon}{\sqrt{2\pi n}} e^{n\left(H(\alpha+\varepsilon)-H\left(\frac{1}{2}\right)\right)}$.

Therefore, $P(\#\text{Heads} \in ((\alpha-\varepsilon)n, (\alpha+\varepsilon)n)) \longrightarrow 0$ exponentially as $n \to \infty$.

Hence,

$$P\left(\#\text{Heads} \in n\left(\frac{1}{2}-\varepsilon, \frac{1}{2}+\varepsilon\right)\right) \longrightarrow 1.$$

This is called the **Large Deviation Phenomenon**.

$P(\text{staying close to mean}) \longrightarrow 1$,

whereas $P(\text{deviating from the mean significantly}) \longrightarrow 0$ exponentially.

In our case, the rate function is $H(\alpha) - H\left(\frac{1}{2}\right)$.

Let $f_n = \frac{\#\text{Heads}}{n}$. Then $f_n \longrightarrow f \qquad$ in measure.

Therefore,

$$P\left(\left|\frac{\#\text{Heads}}{n} - \frac{1}{2}\right| > \varepsilon\right) \longrightarrow 0 \qquad \text{as } n \to \infty. \tag{1}$$

We call this **convergence in probability**.

Later we shall use the **Borel–Cantelli Lemma** to prove from (1) that $\frac{\#\text{Heads}}{n} \longrightarrow \frac{1}{2}$ almost surely.

The sample space is $\Omega = \{H, T\}^{\mathbb{N}}$. We equip Ω with the product topology, where each copy of

$\{H, T\}$ carries the discrete topology. The corresponding Borel σ-algebra, $\mathcal{F} = \mathcal{B}(\Omega)$, is generated by the cylinder sets. The basis of the product topology consists of the cylinder sets, namely sets of the form $\{\omega \in \Omega : \omega_{i_1} = a_1, \ldots, \omega_{i_k} = a_k\}$, where $i_1, \ldots, i_k$ are finitely many coordinates and each $a_j \in \{H, T\}$.

If we only specify the first n coordinates, they look like $\{\omega : \omega_1 = a_1, \ldots, \omega_n = a_n\}$, which are the basic cylinders.

The complete probability space is $(\Omega, \mathcal{F}, \mu) = \left(\{H, T\}^{\mathbb{N}}, \mathcal{B}\left(\{H, T\}^{\mathbb{N}}\right), \mu\right)$,

Recall,

$$P\left(\#\text{Heads} = \frac{n}{2}\right) \sim \frac{2}{\sqrt{2\pi n}}.$$

Suppose

$$c > 0,$$

and let $\left|k - \frac{n}{2}\right| < c\sqrt{n}$.

Then

$$P(\#\text{Heads} = k) = \frac{\frac{n!}{(n-k)!k!}}{2^n}.$$

Using Stirling's formula,

$$P(\#\text{Heads} = k) \sim \frac{1}{\sqrt{\pi n/2}} \cdot \frac{1}{\left(2 - \frac{2k}{n}\right)^{n-k} \left(\frac{2k}{n}\right)^{k}}.$$

$$= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \cdot \frac{1}{\left(1 + \left(1 - \frac{2k}{n}\right)\right)^{n-k} \left(1 + \left(\frac{2k}{n} - 1\right)\right)^{k}}.$$

Hence,

$$= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \exp\left(-\frac{(n-2k)^2}{n}\right).$$

Equivalently, $= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \exp\left(-\frac{\left(k - \frac{n}{2}\right)^2}{2 \cdot \frac{n}{4}}\right)$, which resembles the normal distribution.

$$\mu = \frac{1}{2}, \qquad \sigma = \frac{1}{2}.$$

Hence,

$$= \frac{1}{\sqrt{2\pi\sigma^2 n}} \exp\left(-\frac{(k-\mu n)^2}{2\sigma^2 n}\right).$$

We can prove, $P\left(a < \frac{\#\text{Heads} - \mu n}{\sigma\sqrt{n}} < b\right) \longrightarrow \int_a^b \varphi(x)\, dx$, as $n \to \infty$. (**Central Limit Theorem / De Moivre–Laplace**)

End of Lecture

2.2 Quiz, 10 August 2026

Quiz 1

1. Let $p < \frac{1}{2}$. Suppose we toss n coins independently such that the probability of heads is p and tails is $q = 1 - p$.

   For what value of k is the probability of getting k heads maximised?

2. For probability vectors $P = (p_1, p_2, \ldots, p_n), \qquad Q = (q_1, q_2, \ldots, q_n)$, where $p_i, q_i > 0, \qquad \sum_i p_i = \sum_i q_i = 1$, define the Kullback–Leibler divergence by

$$D_{KL}(P \| Q) = \sum_i p_i \log \left(\frac{p_i}{q_i} \right).$$

   Use Jensen's inequality to show that $D_{KL}(P \| Q) \geq 0$ with equality if and only if $P = Q$.

3. Let $\alpha \in (0, 1)$. Prove that the number of heads equalling $n\alpha$ is asymptotically

$$\frac{1}{\sqrt{2\pi n\alpha(1 - \alpha)}} \exp\left(-nD_{KL}(P \| Q)\right),$$

   where $P = (\alpha, 1 - \alpha), \qquad Q = (p, q)$.

Feedback and Corrections

These notes are prepared as lecture notes and may contain typographical errors, omissions, or mathematical mistakes. If you notice any such errors or have suggestions for improving the notes, please feel free to let me know.

Your feedback is greatly appreciated and will help improve future versions of these notes.

Report an error or suggest a correction

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